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B.Tech. Degree VI Semester Examination April 2016

ME 1603 OPERATIONS MANAGEMENT (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain critical activities, non-critical activities and dummy activities.
 (b) Differentiate moving average and exponential smoothing forecasting methods.
 (c) Enumerate the important qualities in the design of a product.
 (d) Describe inventory control and the relevant costs.
 (e) What are the variables in aggregate planning?
 (f) Explain the critical ratio scheduling method.
 (g) What are the objectives of plant layout?
 (h) Explain depreciation.

PART B

(4 × 15 = 60)

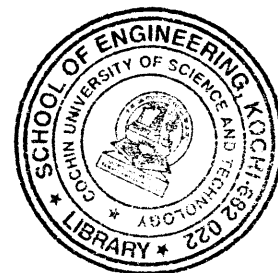
- II. A project consists of the activities as given in the table below.

(15)

Activity	Optimistic time (weeks)	Most likely time (weeks)	Pessimistic time (weeks)
1-2	1	2	3
2-3	1	2	3
2-4	1	3	5
3-5	3	4	5
4-5	2	3	4
4-6	3	5	7
5-7	4	5	6
6-7	6	7	8
7-8	2	4	6
7-9	4	6	8
8-10	1	2	3
9-10	3	5	7

- (i) Draw the project network.
 (ii) Find the expected duration and variance of the activities.
 (iii) Find expected duration of the project.
 (iv) Find the probability of completing the project within 30 weeks.

OR



- III. A firm use simple exponential smoothing with $\alpha=0.2$ to forecast demand. The actual demand is as shown in the table below. (15)

Month	1	2	3	4	5	6	7
Demand	1450	1460	1465	1434	1420	1498	1462

Forecast for the first month was 1400 units. Forecast the demand for months 2 to 7.

- IV. Explain the various factors to be considered under product analysis. (15)

OR

- V. What is EOQ? What are the limitations and assumptions? (15)

- VI. Find the sequence that minimize the total elapsed time required to complete the jobs on two machines M1 and M2 in the order M1M2. The processing time in hours is as given below. (15)

Job	A	B	C	D	E
M1	50	10	90	30	100
M2	20	60	70	80	40

OR

- VII. (a) Discuss on aggregate planning strategies. (5)
(b) Sequence the jobs waiting to be processed on 1st January in a machine shop using critical ratio method for the data given below. (10)

Job	1	2	3	4	5
Processing time (days)	6	2	3	4	5
Due date (January)	18	6	9	11	8

- VIII. Discuss on the factors influencing plant location. (15)

OR

- IX. Describe the economic aspects of maintenance. How can an optimum maintenance plan be selected? (15)

